

Newton's Laws of Motion

First Law: An object at rest stays at rest, and an object in motion stays in motion at constant velocity, unless acted on by a net external force. This is also called the law of inertia.

Second Law: The acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass. This is written as $F = m \times a$.

Third Law: For every action there is an equal and opposite reaction. When object A exerts a force on object B, object B exerts an equal and opposite force back on object A.